Math 420: Homework 1

Name:

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1. For any square matrix A and any integer $k \geq 4$, prove the identity

$$A^k - I = (A - I)(A^{k-1} + A^{k-2} + \cdots + A + I).$$

2. Let A be a 3×3 matrix that commutes with all 3×3 matrices. Prove that A is a scalar multiple of the identity matrix.

3. Prove that the set $V = \{(x, y, z) \in \mathbb{R}^3 : z = 3x \text{ and } y = 2z\}$ is a vector space.

4. Show that the following are not vector spaces

   (a) $V = \{(x, y, z) \in \mathbb{R}^3 : z = xy\}$

   (b) $V = \{(x, y, z) \in \mathbb{Z}^3 : z = x + y\}$

   (c) V is the set of 3×3 matrices with real coefficients where vector addition is defined to be matrix multiplication.

5. Explain how $\mathbb{C}^n$ can be thought of as a vector space over $\mathbb{R}$. How many real numbers would be needed for this vector space?

6. Let V be a vector space with subspaces U and W. Prove that

$$U \cup W = \{v \in V : v \in U \text{ or } v \in W\}$$

is a subspace of V if and only if either $U \leq W$ or $W \leq U$.